

Computer Abstraction and Technology.

May 24, 2026

1.1 [2] <§1.1> Aside from the smart cell phones used by a billion people, list and describe four other types of computers.

- (a) Personal computers. They emphasize on good performance to a single user at a low cost. They usually execute third party software.
- (b) Embedded computers. The largest classes of computers. They are designed to run one application that's usually integrated with the hardware and delivered as a single system.
- (c) Servers Usually accessed via network. Servers are oriented towards carrying a sizable complex workload which may be a single application that's encompasses a scientific application or an engineering application - or it may handle multiple small jobs. Servers are built from the same technology as desktop computers but provide for greater computing, storage, and input/output capacity. Servers place more emphasis on dependability. Servers span the widest range in terms of cost and capability.
- (d) Supercomputers. Consist of thousands to millions of processors and terabytes of storage. They are used for high-end scientific and engineering calculations, such as weather forecasting, oil exploration, protein structure determination and other large scale problems.

1.2 [5] <§1.2> The eight great ideas in computer architecture are similar to ideas from other fields. Match the eight ideas from computer architecture, "Design for Moore's Law," "Use Abstraction to Simplify Design," "Make the Common Case

Fast,” “Performance via Parallelism,” “Performance via Pipelining,” “Performance via Prediction,” “Hierarchy of Memories,” and “Dependability via Redundancy” to the following ideas from other fields:

- (a) Assembly lines in automobile manufacturing
Performance via Pipelining
- (b) Suspension bridge cables
Performance via Parallelism
- (c) Aircraft and marine navigation systems that incorporate wind information
Use Abstraction to Simplify Design
- (d) Express elevators in buildings
Make the Common Case Fast
- (e) Library reserve desk
Hierarchy of Memories
- (f) Increasing the gate area on a CMOS transistor to decrease its switching time
Design for Moore’s Law
- (g) Adding electromagnetic aircraft catapults (which are electrically powered as opposed to current steam-powered models), allowed by the increased power generation offered by the new reactor technology.
Performance via Prediction
- (h) Building self-driving cars whose control systems partially rely on existing sensor systems already installed into the base vehicle, such as lane departure systems and smart cruise control systems.
Dependability via Redundancy

1.3 [2]<§1.3> Describe the steps that transform a program written in a high-level language such as C into a representation that is directly executed by a computer processor.

- (a) High-level language -> Compiled to -> Assembly language
-> Assembler(Assembles the Assembly language to) -> Machine language(Binary instructions)